

length of string

1st call

len_string("hello1").

return 1 + len_string("ello1") $\Rightarrow 1 + 5 = 6$

2nd call

len_string("ello1")

return 1 + len_string("llo1") $\Rightarrow 1 + 4 = 5$

3rd call

len_string("llo1")

return 1 + len_string("lo1") $\Rightarrow 1 + 3 = 4$

4th call

len_string("lo1")

return 1 + len_string("o1") $\Rightarrow 1 + 2 = 3$

5th call

len_string("o1")

return 1 + len_string("1") $\Rightarrow 1 + 1 = 2$

6th call

len_string("1")

return 1 + len_string("") $\Rightarrow 1 + 0$

Factorial of 5.

1st call.

factorial(5).

return = $5 * \text{factorial}(4)$. $\Rightarrow 5 \times 24 = 120$

2nd call

factorial(4)

return $4 * \text{fact}(3)$. $\Rightarrow 4 \times 6 = 24$

3rd call

fact(3)

return $3 * \text{fact}(2)$. $\Rightarrow 3 \times 2 = 6$

4th call

fact(2)

return $2 * \text{fact}(1)$. $\Rightarrow 2 \times 1 = 2$

5th call

fact(1)

return 1